Release Notes & Technical Reference Guide

Official Release Documentation for General Availability v1.0.0 | Production Build | ISO 14289-1 (PDF/UA-1) & WCAG 2.1 / 2.2 AA Standard

Release Version:	v1.0.0 (General Availability)	**Distribution Target:**	Windows 10 / 11 64-bit (Native x64)
Release Classification:	Major Production Release	**Packaging Format:**	Single-File Standalone Portable EXE
Supported Standards:	PDF/UA-1, Matterhorn 1.1, WCAG 2.1/2.2 AA	**Runtime Requirements:**	Zero External Dependencies (Self-Contained)
Core Parser Engine:	Dual Engine: pikepdf (QPDF) + PyMuPDF (MuPDF)	**Reporting Formats:**	PDF Certificate, HTML, JSON, CSV

1. Executive Overview

PDF Accessibility Inspector v1.0.0 delivers an enterprise-grade, independent desktop auditing solution engineered for document authors, accessibility specialists, compliance officers, and remediation teams. Combining the high-performance C++ core of QPDF (via pikepdf) with the rapid layout rendering and geometry calculation of MuPDF (via PyMuPDF), this release introduces deep structural validation, interactive visual element tracking, simulated screen reader playback, and audit reporting.

2. Standards & Checkpoint Coverage

Standard / Specification	Scope & Evaluation Focus	Coverage Level
PDF/UA-1 (ISO 14289-1:2014)	Audits catalog mark flags, logical structure tree root, marked content (/MCID), font embedding, /ToUnicode CMaps, natural language tags (IETF BCP 47), alternate descriptions for figures, table header symmetry, and XMP metadata packet validity.	**Full Parity (Automated)**
Matterhorn Protocol 1.1	Evaluates machine-testable failure conditions across 31 checkpoint items encompassing structure elements, role mapping loops, artifacts nesting, table headers, and form field descriptors.	**31 Checkpoints Automated**
WCAG 2.1 & 2.2 (Level A & AA)	Evaluates all 4 core principles: Perceivable (1.1.1 Non-text, 1.3.1 Info & Relationships, 1.4.3 Contrast), Operable (2.1.1 Keyboard Order, 2.4.2 Page Title, 2.4.5 Bookmarks, 2.4.4 Link Purpose), Understandable (3.1.1/3.1.2 Language, 3.3.2 Form Labels), and Robust (4.1.2 Name/Role/Value).	**Comprehensive A & AA Suites**
Ergonomics & Quality	Analyzes heading level progression (detects skipped tiers like H1 to H3), table matrix regularity, ambiguous link text ('click here', 'learn more'), bare URLs, empty structural containers, and screen reader flow continuity.	**Deep Heuristics & Quality Scoring**

3. Major Features & Technical Capabilities in v1.0.0

• **Dual-Engine Deep PDF Structural Inspection**: Integrates pikepdf (QPDF C++ core) for low-level PDF dictionary, catalog, and StructTree analysis with PyMuPDF (MuPDF C core) for fast rendering, text flow analysis, and visual coordinate mapping. Validates /StructTreeRoot, /ParentTree, Marked Content IDs (/MCID), /RoleMap custom roles, font embedding, /ToUnicode CMaps, /ViewerPreferences, and XMP metadata.

• **Interactive PAC Matrix & Category Drill-Down**: Presents compliance results in a structured matrix categorized by ISO and WCAG checkpoint groups. Provides interactive status filtering, immediate category breakdown in the right inspection panel, and handling for NOT_APPLICABLE status checkpoints with explanatory badges.

• **WCAG 2.1 & 2.2 Success Criteria Reference Guide**: Directly incorporates official W3C Success Criteria guidance into the UI details panel and checkpoint hover tooltips. Includes normative criteria definitions, intent summaries, and remediation recommendations for PDF

authoring tools (Microsoft Word, Adobe InDesign, Adobe Acrobat Pro).

• **Click-to-Highlight Visual Element Locator**: Synchronized PDF canvas that jumps directly to the affected page and illuminates offending elements with animated, colored bounding box overlays upon clicking any finding in the audit tables.

• **Interactive Tag Tree Explorer**: Hierarchical tree view mapping the physical structure of the PDF from /StructTreeRoot down to content items. Displays resolved standard roles, custom tags, titles, attributes, language overrides, and alternate descriptions for any selected tag.

• **Simulated Screen Reader Preview (Semantic Linearizer)**: Linearizes document content according to the logical structure tree reading sequence, providing an auditory preview of how screen readers (NVDA, JAWS, VoiceOver) speak headings, lists, tables, links, and alternative descriptions.

• **Asset & Element Extractors**: Dedicated inspector views for granular asset inventory: Fonts (embedding status, subset flags, /ToUnicode), Images & Figures (dimensions, bounding boxes, /Alt text presence), Tables (matrix regularity, headers, row/col spans), Links (URIs, anchor text clarity), Form Fields (tooltips /TU, field names /T), and Bookmarks.

• **Multi-File Batch Folder Scanner**: Scans entire directories of PDF documents using background multi-threading. Displays aggregate compliance scores, pass/warning/fail tallies, and enables single-click loading into the primary inspection workspace.

• **Publication-Grade Multi-Format Reporting**: Generates publication-quality reports across four distinct formats: formal PDF compliance certificates (via ReportLab), interactive standalone HTML reports with dynamic charts and filters, and machine-readable JSON and CSV exports for CI/CD integration.

• **Password-Protected Document Support**: Secure, in-memory credential unlocking allows seamless inspection of encrypted PDF files without persisting credentials to disk.

4. Engineering Hardening & Bug Fixes in v1.0.0

• **Marked Content ID (/MCID) Resolution Fix**: Corrected deep dictionary traversals where structure element marked content identifiers were previously miscalculated in complex nested grouping tags (Sect, Part, Art).

• **Accurate Zero-Applicable Checkpoint Handling**: Implemented formal NOT_APPLICABLE badge and status resolution so checkpoints that do not apply to a specific document (e.g. form field rules on a book without forms) are clearly explained rather than falsely marked as pass or fail.

• **Heading Level Skip Detection Refinement**: Enhanced QUAL-HEAD-001 to accurately identify improper tier jumps (e.g. H1 directly to H3 or H4) without generating false positives on nested sub-sections.

• **UI Layout & Off-Screen Protection**: Implemented automatic screen-aware geometry clamping ensuring the main application window and floating dialogs never launch off-screen or collapse splitter panels on multi-monitor or high-DPI setups.

• **Memory Safety & Background Threading**: Offloaded PDF parsing, structure tree traversal, and rule execution to dedicated QThread workers, maintaining a responsive 60 FPS UI during audits of 500+ page documents.

• **Packaging & PyInstaller Hardening**: Standardized asset path resolution for stylesheets (theme.qss), icons, and native C++ dynamic libraries (pikepdf._qpdf, fitz), ensuring clean single-file executable execution on clean Windows environments.

5. System Requirements & Distribution

Component	Requirement / Target Specification
Operating System	Windows 10 / Windows 11 (64-bit Architecture)
Pre-Built Executable	dist/PDF-Accessibility-Inspector.exe (Standalone, No Python Required)
Development Runtime	Python 3.10, 3.11, or 3.12 (64-bit)
Core Libraries	PySide6 >= 6.6.0, PyMuPDF >= 1.24.0, pikepdf >= 8.0.0, ReportLab >= 4.0.0, Pillow >= 10.0.0
Hardware Specs	Minimum 4 GB RAM (8 GB recommended for 500+ page PDFs), 250 MB free disk space

6. Quick Start & Execution

Running from Standalone Executable: Double-click dist/PDF-Accessibility-Inspector.exe to launch.
Running from Source (PowerShell):
pip install -r requirements.txt
python run.py
Running Test Suite & Parity Audits:
pytest -v
python scripts/verify_all_pdf_types.py